SEPARATING GASES FROM THE ATMOSPHERE 3

The gases that surround Earth are known as its atmosphere. The atmosphere is a source of many important gases. These gases can be obtained from the air by **fractional distillation**. In this process, air is compressed and cooled until it is a liquid. The temperature is gradually increased and gases are distilled off in the order of their boiling points.

The gas with the lowest boiling point will be the first to vaporise and be distilled off. As the temperature continues to rise, the gas with the next lowest boiling point will vaporise, and so on. The gas with the highest boiling point will be the last to be distilled. Figure 1 shows a simplified version of how two gases separate during this process.



Figure 1

1 Define *boiling point*.

Liquid nitrogen has a lower boiling point than liquid oxygen. This means that in fractional distillation it distils off before oxygen does. Liquid nitrogen is very cold (–196°C) and can be used to freeze food very quickly (flash-freezing) or to keep food frozen during transportation. Nitrogen is a colourless, odourless, tasteless, non-toxic gas. It is very unreactive and is sometimes used to fill space in food packages such as potato chips to prevent oxidation and physical damage. It is also used in oil tanks to reduce the risk of fire.

2 Why is food frozen? Name one item other than food that could be flash-frozen.

3 Explain what is meant by *non-toxic*.

4 Define *cryogenic* and *superconductivity*.

Oxygen is also obtained from liquid air by fractional distillation. Oxygen is a clear, colourless gas with no odour. It is slightly soluble in water (i.e. small amounts can dissolve in water). Unlike nitrogen, it is very reactive and combines with many substances. Oxygen is compressed and put into cylinders for use by people who go where there is little or no air; e.g. people in high-altitude aircraft, firefighters, astronauts and scuba divers. It is also used in medical situations where people have breathing difficulties; e.g. oxygen tents are placed around people so that they can breathe in pure oxygen.

The **'noble gases'** (helium, neon, argon, krypton and xenon) can also be extracted from the air. They are all very unreactive substances. Helium (commercially extracted from methane) is used for lighter-than-air balloons (blimps/dirigibles) and mixed with oxygen to make a breathing mixture for scuba divers. Helium occurs as a liquid just above absolute zero (–273°C) and has a boiling point of –269°C. It can be used to cool substances close to absolute zero and is used in experiments in superconductivity and cryogenics (the study and use of substances at very low temperatures).

Neon can also be used as a cryogenic refrigerant but is more commonly applied in 'neon lighting', where electric discharge tubes used in advertising signs give a crimson colour.

Argon is used to fill incandescent filament light globes. It prevents the tungsten filament from burning. It also gives a blue light in neon lighting and is used in electric welding to prevent the metal being welded from burning. Krypton gives a bright orange-red emission in neon lighting.

5 Draw up a table giving the names, symbols, properties and uses of the gases mentioned on this page.

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978 1 92561 065 9